

Tutorial 1

Concepts Tested

Variables, operator, program compilation, program structure

1. Write a simple calculator C++ program to accept 2 integer number x1 and x2 and perform the following operation on the two values. The result should be displayed in the format as shown at <https://www.calculator.net/>
Test your calculator with a few examples that include the values (8, 5), (9,8)
Compile your program with your graphical integrated development tool (IDE) such as Code Block. Try to compile the program using command line compiler g++ , for example
g++ program1.cpp -o prog1
The above command produces the exe file prog1.exe

- i) Addition
- ii) Subtraction
- iii) Multiplication
- iv) Division
- v) Remainder

Example program flow

```
Simple calculator
Enter option: 1)+ 2)- 3)* 4)/ 5) mod(x1,x2) -1) Exit  Your selection -> 1

Enter first number : 1

Enter second number : 2

1 + 2 = 3
Press any key to continue . . .

Simple calculator
Enter option: 1)+ 2)- 3)* 4)/ 5) mod(x1,x2) -1) Exit  Your selection -> -1
```

2. Write a C++ program to calculate your body mass index (BMI). Follow the formula below and display the weight categories as stated at https://www.nhlbi.nih.gov/health/educational/lose_wt/BMI/bmi-m.htm

$$\text{BMI} = \frac{m}{h^2}$$

BMI = body mass index

m = mass (in kilograms)

h = height (in meters)

Get weight in kg :75

Get height in meter :1.65

Your bmi = 27.5482

Your weight category is : Overweight